

Qualification Pack



Mechanic -Fitter (Mineral Sands)

QP Code: MIN/Q3211

Version: 1.0

NSQF Level: 3.0

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MIN/Q3211: Mechanic -Fitter (Mineral Sands)

Brief Job Description

A Mechanic Fitter (Mineral Sands) is responsible for the maintenance and repair of different types of processing and material handling equipment used in dredging and SMPs unit, mineral separation plant, and rare earth extraction plant.

Personal Attributes

The individual should have attention to detail, organizational and problem-solving skills. The person should also have appropriate verbal and written communication skills.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [MIN/N3224: Maintain and repair the dredging and SMPs unit equipment](#)
2. [MIN/N3225: Maintain and repair the mineral separation plant equipment](#)
3. [MIN/N3226: Maintain and repair the rare earth extraction plant equipment](#)
4. [MIN/N1810: Follow the applicable health, safety and environmental guidelines \(Placer Mining\)](#)
5. [MIN/N1811: Work collaboratively and effectively in a team](#)
6. [DGT/VSQ/N0101: Employability Skills \(30 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Mining
Sub-Sector	Engineering Services
Occupation	Mechanical Services
Country	India
NSQF Level	3.0
Credits	12
Aligned to NCO/ISCO/ISIC Code	NCO-2015/ 3115.9900, NCO-2015/ 2146.9900

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Minimum Educational Qualification & Experience	10th grade pass OR 8th grade pass (with two year of NTC/NAC after 8th) OR 8th grade pass (with 2 years of relevant experience) OR 5th grade pass (with 5 years of relevant experience) OR Previous relevant Qualification of NSQF Level (2)(with 3 years of relevant experience) OR Previous relevant Qualification of NSQF Level (2.5)(with 1.5 year of relevant experience)
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	14/03/2027
NSQC Approval Date	15/03/2024
Version	1.0
Reference code on NQR	QG-03-MG-02171-2024-V1-SCMS
NQR Version	1.0

Remarks:

360 Hours Duration

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MIN/N3224: Maintain and repair the dredging and SMPs unit equipment

Description

This occupational standards unit is about carrying out the dredging operation.

Scope

The scope covers the following :

- Maintain the dredge
- Maintain the trommel
- Maintain the spiral
- Maintain the static and vibrating screen
- Maintain the HEMM

Elements and Performance Criteria

Maintain the dredge

To be competent, the user/individual on the job must be able to:

- PC1.** follow the manufacturer's instructions for the safe handling and maintenance of the dredge
- PC2.** carry out maintenance of hydraulic system, including power pack, actuators, direction control valves, pressure control valve, check valve, hoses, coupling, etc.
- PC3.** check the operating parameters of the dredger unit, e.g. main motor temperature, mass flow meter reading, main motor ammeter reading, head feed bin level indicator reading etc., are within the permissible limit
- PC4.** check for the correct functioning of the slurry pump
- PC5.** monitor the pumps for abnormal noise, vibration, overheating of bearing, loose belt, etc
- PC6.** carry out maintenance of the pumps, including belt tension, foundation bolt, pulley replacement, alignment of belt, greasing and lubrication
- PC7.** assemble and dismantle pumps and pipelines
- PC8.** use the correct components/ spares and consumable for the maintenance

Maintain the trommel

To be competent, the user/individual on the job must be able to:

- PC9.** check for the correct functioning of trommel components, e.g. drive mechanism, rollers, guards, screen, trommel tires, feed end, discharge chute, roller bearings, fusible plug in fluid coupling, etc
- PC10.** maintain the cleanliness of the screen in the trommel to prevent any blockage/ jamming
- PC11.** check different parameters, such as ampere meter reading of drive motor, feed rate, discharge pressure, head feed bin level indicator, and mass flow meter reading
- PC12.** check the condition of support roller, screens, gearbox, inlet chute, pulley alignment, belt drive, and roller bearing carry out greasing and lubrication of the moving parts
- PC13.** identify issues such as vibration and unusual noises in the trommel and take appropriate corrective measures

Maintain the Spiral

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To be competent, the user/individual on the job must be able to:

PC14. check for the correct functioning of spiral

PC15. ensure the pumps connected to spiral circuits are well-maintained

Maintain the static and vibrating screen

To be competent, the user/individual on the job must be able to:

PC16. check for the correct functioning and installation of static and vibrating screen

PC17. check for vibration and abnormal noises and take appropriate corrective measures

PC18. examine the alignment of the screen and adjust it as required

PC19. check the condition of the bearing unit, coupling, screen mesh, feeding hopper, and frame, and repair or replace them as appropriate

PC20. identify the need for and perform greasing and lubrication of the moving parts

PC21. use the appropriate digital diagnostic tools to assess the issues with the static and vibrating screen

PC22. use the pressure gauges, filler gauges, callipers and condition monitoring equipment to detect faults in the system

Maintain the HEMM

To be competent, the user/individual on the job must be able to:

PC23. follow the appropriate methods to assess the vehicle to identify damage, corrosion, fluid levels, leaks, etc.

PC24. conduct routine examinations as per the maintenance schedule recommended by the manufacturer

PC25. check the lubricant level and refill as required, using the lubricant of appropriate grades

PC26. follow the OEM-recommended procedure and checklist for routine servicing of HEMM

PC27. identify wear and tear of oil and air filters, drive belts, and braking system, and replace them as required

PC28. carry out breakdown maintenance in the workshop or work site when required

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. different dredging equipment used in the beach sand mining industry

KU2. the working principle of the slurry pump

KU3. the functions of different components of the slurry pump and the pump characteristic/performance curves

KU4. different types of Heavy Earth Moving Machinery (HEMM)

KU5. the basic technology used in and functioning of various components and equipment in the dredging process

KU6. the use of appropriate tools to assess and confirm technical faults

KU7. the use of instruments like pressure indicators, fuel pressure testers, manifold gauge sets, oil pressure gauges, measuring equipment (Vernier callipers, micrometer, feeler gauges, multi-meter, flow meter, temp gauge, dial gauge etc.), condition monitoring systems for temperature, vibration, sound, etc.

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- KU8.** Indian rare earth mining rules and regulations
- KU9.** the appropriate safety measures to be taken in the vicinity of machinery
- KU10.** the importance of following the locally prepared emergency response / disaster management plan
- KU11.** the importance of promptly resolving or reporting the issues identified with the equipment
- KU12.** the consequences and cost to the organization due to accidents/ delays and in-operational equipment
- KU13.** the importance of being aware of the layout of the work area
- KU14.** the benefits of following the daily maintenance checklist
- KU15.** the role of workmen under the Mines Act
- KU16.** the implications of violating the safety procedures
- KU17.** the appropriate precautions to be taken when handling compressors, pumps, dryers, spirals, trommel
- KU18.** the use of appropriate hand and power tools
- KU19.** the importance and use of different grades of lubricants
- KU20.** the use of different types of nuts, bolts, seals and couplings
- KU21.** the do's and don'ts of equipment repair and maintenance
- KU22.** the maintenance of hydraulic systems, including different types of pumps and control valves
- KU23.** the safety precautions for using tools and tackling machine parts
- KU24.** the SOP for safe operation and maintenance of plant
- KU25.** the typical causes and symptoms of operational faults and failures of a machine
- KU26.** the appropriate corrective and preventive measures for machine faults
- KU27.** the disposal of replaced components in accordance with the safety, health and environmental policies and regulations
- KU28.** the assembling techniques for different types of bearings used in machines
- KU29.** the standard procedure for performing preventive maintenance
- KU30.** the appropriate fitting and assembly techniques
- KU31.** the importance of following the appropriate safety practices to ensure own and others' safety

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** make note of work-related observations
- GS2.** interpret the relevant drawings, sketches, symbols, measurements and Original Equipment Manufacturer (OEM) manuals
- GS3.** use the online Enterprise Resource Planning (ERP) systems under guidance of the supervisor
- GS4.** communicate politely and professionally
- GS5.** use effective questioning techniques to extract maximum information about different tasks
- GS6.** plan daily tasks appropriately to ensure their timely completion



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- GS7.** undertake different activities as per the applicable task lists and schedules with minimum supervision
- GS8.** make appropriate decisions concerning the area of work
- GS9.** follow the 5s principles at work
- GS10.** use reasoning skills to identify and resolve basic problems at work
- GS11.** assist in identifying and implementing appropriate solutions to work-related problems
- GS12.** follow the organizational protocol during emergencies

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Maintain the dredge</i>	10	13	-	5
PC1. follow the manufacturer's instructions for the safe handling and maintenance of the dredge	1	1	-	1
PC2. carry out maintenance of hydraulic system, including power pack, actuators, direction control valves, pressure control valve, check valve, hoses, coupling, etc.	2	2	-	1
PC3. check the operating parameters of the dredger unit, e.g. main motor temperature, mass flow meter reading, main motor ammeter reading, head feed bin level indicator reading etc., are within the permissible limit	1	2	-	1
PC4. check for the correct functioning of the slurry pump	1	2	-	-
PC5. monitor the pumps for abnormal noise, vibration, overheating of bearing, loose belt, etc	1	1	-	-
PC6. carry out maintenance of the pumps, including belt tension, foundation bolt, pulley replacement, alignment of belt, greasing and lubrication	2	2	-	1
PC7. assemble and dismantle pumps and pipelines	1	2	-	-
PC8. use the correct components/ spares and consumable for the maintenance	1	1	-	1
<i>Maintain the trommel</i>	5	10	-	4
PC9. check for the correct functioning of trommel components, e.g. drive mechanism, rollers, guards, screen, trommel tires, feed end, discharge chute, roller bearings, fusible plug in fluid coupling, etc	1	2	-	1
PC10. maintain the cleanliness of the screen in the trommel to prevent any blockage/ jamming	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. check different parameters, such as ampere meter reading of drive motor, feed rate, discharge pressure, head feed bin level indicator, and mass flow meter reading	1	2	-	1
PC12. check the condition of support roller, screens, gearbox, inlet chute, pulley alignment, belt drive, and roller bearing carry out greasing and lubrication of the moving parts	1	2	-	1
PC13. identify issues such as vibration and unusual noises in the trommel and take appropriate corrective measures	1	2	-	1
<i>Maintain the Spiral</i>	2	3	-	1
PC14. check for the correct functioning of spiral	1	2	-	1
PC15. ensure the pumps connected to spiral circuits are well-maintained	1	1	-	-
<i>Maintain the static and vibrating screen</i>	7	13	-	6
PC16. check for the correct functioning and installation of static and vibrating screen	1	2	-	1
PC17. check for vibration and abnormal noises and take appropriate corrective measures	1	2	-	1
PC18. examine the alignment of the screen and adjust it as required	1	1	-	-
PC19. check the condition of the bearing unit, coupling, screen mesh, feeding hopper, and frame, and repair or replace them as appropriate	1	2	-	1
PC20. identify the need for and perform greasing and lubrication of the moving parts	1	2	-	1
PC21. use the appropriate digital diagnostic tools to assess the issues with the static and vibrating screen	1	2	-	1
PC22. use the pressure gauges, filler gauges, callipers and condition monitoring equipment to detect faults in the system	1	2	-	1
<i>Maintain the HEMM</i>	6	11	-	4

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC23. follow the appropriate methods to assess the vehicle to identify damage, corrosion, fluid levels, leaks, etc.	1	1	-	1
PC24. conduct routine examinations as per the maintenance schedule recommended by the manufacturer	1	2	-	1
PC25. check the lubricant level and refill as required, using the lubricant of appropriate grades	1	2	-	1
PC26. follow the OEM-recommended procedure and checklist for routine servicing of HEMM	1	2	-	-
PC27. identify wear and tear of oil and air filters, drive belts, and braking system, and replace them as required	1	2	-	1
PC28. carry out breakdown maintenance in the workshop or work site when required	1	2	-	-
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	MIN/N3224
NOS Name	Maintain and repair the dredging and SMPs unit equipment
Sector	Mining
Sub-Sector	Engineering Services
Occupation	Mechanical Services
NSQF Level	3.0
Credits	4
Version	1.0
Last Reviewed Date	15/03/2024
Next Review Date	14/03/2027
NSQC Clearance Date	15/03/2024

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MIN/N3225: Maintain and repair the mineral separation plant equipment

Description

This occupational standards unit is about maintaining and repairing different types of equipment used in an earth extraction plant.

Scope

The scope covers the following :

- Maintain the mineral separation plant equipment
- Conduct preventive maintenance of processing equipment components
- Track the preventive maintenance activities

Elements and Performance Criteria

Maintain the mineral separation plant equipment

To be competent, the user/individual on the job must be able to:

- PC1.** follow the manufacturer's instructions for the safe use and maintenance of the mineral separation plant equipment
- PC2.** check for the correct functioning of different types of separators, e.g. High Tension Separator, Electric plate separator, Magnetic Separator, Induced Roll Magnetic Separator, Lift Roll Magnetic Separator, etc
- PC3.** check for the correct functioning of material handling equipment, e.g. bucket elevator, bucket belt conveyor, belt conveyor, screw feeder, and different c, e.g. rotary drier, fluidized bed drier, and shaft drier
- PC4.** check for the correct functioning of different components of the power transmission systems, e.g. gear and gearboxes, couplings, V belts, V pulleys, etc
- PC5.** follow the appropriate procedure of mounting, assembling, and disassembling different types of bearings
- PC6.** check for the correct functioning of water pumps, slurry pumps, oil pumps, vacuum pumps, and carry out appropriate troubleshooting as required
- PC7.** assist in the installation, repair and maintenance of different types of machinery
- PC8.** use the relevant mechanical tools following the applicable blueprints and manuals

Conduct preventive maintenance of processing equipment components

To be competent, the user/individual on the job must be able to:

- PC9.** create maintenance plans as per the schedule recommended by the equipment manufacturer
- PC10.** perform routine inspections of the manufacturing equipment
- PC11.** inspect valves, linkages, bearings, hydraulic and pneumatic systems for smooth operation
- PC12.** monitor the performance of different plant machinery, detect malfunctions and take appropriate corrective measures
- PC13.** resolve issues with the equipment by applying equipment knowledge and standard procedures

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Track the preventive maintenance activities

To be competent, the user/individual on the job must be able to:

- PC14.** conduct the appropriate quality checks and maintain relevant records as per the standard procedure
- PC15.** identify and report quality issues to the supervisor, following the organizational procedure
- PC16.** undertake appropriate adjustments to the equipment and machinery when necessary
- PC17.** track the hours-in-operation for different equipment and adhere to the applicable preventive maintenance schedules
- PC18.** coordinate with the supervisor to resolve the equipment issues that require manufacturer's assistance

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** different mineral separation plant machines used in beach sand mineral processing
- KU2.** different mineral separation plant machine manufacturer's and the machine specifications
- KU3.** different types of separators, e.g. High Tension Separator, Electric plate separator, Magnetic Separator, Induced Roll Magnetic Separator, Lift Roll Magnetic Separator, Rare earths Drum Magnetic Separator, RERMS, Low intensity magnetic separator, Wet High Intensity magnetic separator, Cross Belt Magnetic Separator, High Intensity Magnetic Separator, Gravity Separator - Density Separator- wet table, Air Table, Hydro sizers, Spiral Separator, vibrating screens
- KU4.** different types of bearing
- KU5.** the applicable safety and quality policies, quality objectives and ISO systems
- KU6.** the appropriate safety measures to be taken in the vicinity of machinery
- KU7.** the importance of following the locally prepared emergency response / disaster management plan
- KU8.** the importance of promptly resolving or reporting the issues identified with the equipment
- KU9.** the consequences and cost to the organization due to accidents/ delays and in-operational equipment
- KU10.** the importance of being aware of the layout of the work area
- KU11.** the benefits of following the daily maintenance checklist
- KU12.** the role of workmen under the Mines Act
- KU13.** the implications of violating safety procedures
- KU14.** the appropriate precautions to be taken when handling compressor, pumps, dryers, spirals, trommel
- KU15.** the use of appropriate hand and power tools
- KU16.** the importance and use of different grades of lubricants
- KU17.** the use of different types of nuts, bolts, seals and couplings
- KU18.** the do's and don'ts of equipment repair and maintenance
- KU19.** the maintenance of hydraulic systems, including different types of pumps and control valves
- KU20.** the safety precautions for using tools and tackling machine parts

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- KU21.** the SOP for safe operation and maintenance of plant
- KU22.** the typical causes and symptoms of operational faults and failures of a machine
- KU23.** the appropriate corrective and preventive measures for machine faults
- KU24.** the disposal of replaced components in accordance with the safety, health and environmental policies and regulations

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** make note of work-related observations
- GS2.** interpret the relevant drawings, sketches, symbols, measurements and Original Equipment Manufacturer (OEM) manuals
- GS3.** use the online Enterprise Resource Planning (ERP) systems under guidance of the supervisor
- GS4.** communicate politely and professionally
- GS5.** use effective questioning techniques to extract maximum information about different tasks
- GS6.** plan daily tasks appropriately to ensure their timely completion
- GS7.** undertake different activities as per the applicable task lists and schedules with minimum supervision
- GS8.** make appropriate decisions concerning the area of work
- GS9.** follow the 5s principles at work
- GS10.** use reasoning skills to identify and resolve basic problems at work
- GS11.** assist in identifying and implementing appropriate solutions to work-related problems
- GS12.** follow the organizational protocol during emergencies

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Maintain the mineral separation plant equipment</i>	17	23	-	10
PC1. follow the manufacturer's instructions for the safe use and maintenance of the mineral separation plant equipment	1	3	-	1
PC2. check for the correct functioning of different types of separators, e.g. High Tension Separator, Electric plate separator, Magnetic Separator, Induced Roll Magnetic Separator, Lift Roll Magnetic Separator, etc	3	3	-	2
PC3. check for the correct functioning of material handling equipment, e.g. bucket elevator, bucket belt conveyor, belt conveyor, screw feeder, and different c, e.g. rotary drier, fluidized bed drier, and shaft drier	3	3	-	1
PC4. check for the correct functioning of different components of the power transmission systems, e.g. gear and gearboxes, couplings, V belts, V pulleys, etc	3	3	-	2
PC5. follow the appropriate procedure of mounting, assembling, and disassembling different types of bearings	1	2	-	1
PC6. check for the correct functioning of water pumps, slurry pumps, oil pumps, vacuum pumps, and carry out appropriate troubleshooting as required	3	3	-	1
PC7. assist in the installation, repair and maintenance of different types of machinery	2	3	-	1
PC8. use the relevant mechanical tools following the applicable blueprints and manuals	1	3	-	1
<i>Conduct preventive maintenance of processing equipment components</i>	6	15	-	5
PC9. create maintenance plans as per the schedule recommended by the equipment manufacturer	1	3	-	1

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. perform routine inspections of the manufacturing equipment	1	3	-	1
PC11. inspect valves, linkages, bearings, hydraulic and pneumatic systems for smooth operation	2	3	-	1
PC12. monitor the performance of different plant machinery, detect malfunctions and take appropriate corrective measures	1	3	-	1
PC13. resolve issues with the equipment by applying equipment knowledge and standard procedures	1	3	-	1
<i>Track the preventive maintenance activities</i>	7	12	-	5
PC14. conduct the appropriate quality checks and maintain relevant records as per the standard procedure	2	3	-	1
PC15. identify and report quality issues to the supervisor, following the organizational procedure	2	2	-	1
PC16. undertake appropriate adjustments to the equipment and machinery when necessary	1	2	-	1
PC17. track the hours-in-operation for different equipment and adhere to the applicable preventive maintenance schedules	1	3	-	1
PC18. coordinate with the supervisor to resolve the equipment issues that require manufacturer's assistance	1	2	-	1
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	MIN/N3225
NOS Name	Maintain and repair the mineral separation plant equipment
Sector	Mining
Sub-Sector	Engineering Services
Occupation	Mechanical Services
NSQF Level	3.0
Credits	3
Version	1.0
Last Reviewed Date	15/03/2024
Next Review Date	14/03/2027
NSQC Clearance Date	15/03/2024

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MIN/N3226: Maintain and repair the rare earth extraction plant equipment

Description

This unit is about maintaining and repairing different types of equipment used in an earth extraction plant

Scope

The scope covers the following :

- Assist in installing processing machinery and conveying equipment
- Maintain the rare earth extraction plant equipment

Elements and Performance Criteria

Assist in installing processing machinery and conveying equipment

To be competent, the user/individual on the job must be able to:

- PC1.** prepare the mechanical components for the maintenance of rare earth extraction plant equipment, e.g. grinding, crystallization, evaporators, centrifuge, flash drier setups, filter press, mixer settlers, etc
- PC2.** assemble the relevant equipment in chemical plant processing, material handling, etc

Maintain the rare earth extraction plant equipment

To be competent, the user/individual on the job must be able to:

- PC3.** diagnose and repair faults in mechanical systems, such as gears, pumps, agitators, valves, mixer settler fans, etc
- PC4.** ensure proper protective enclosures are provided for rotating parts, such as drives, belts etc
- PC5.** follow the appropriate safety measures, not performing maintenance of a working machine
- PC6.** conduct appropriate tests (e.g. load testing, vibration testing, etc.) on the equipment after repair and maintenance to ensure everything is working correctly and safely
- PC7.** carry out predictive, preventive and breakdown maintenance for the processing, conveying and other equipment, such as fans, pumps, valves pipelines, gearbox, etc.
- PC8.** follow the maintenance schedule and standard procedure for equipment maintenance
- PC9.** conduct the rubber line spark test

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** different types of unit operations in a rare earth extraction plant
- KU2.** the applicable safety measures and regulations and consequences of violating safety measures
- KU3.** the direct/indirect costs of accidents to the organization
- KU4.** the importance of following the locally prepared emergency response / disaster management plan

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- KU5.** the cost of equipment and loss to the organization due to un-operational equipment
- KU6.** the provision of compensation and working hours as per the Mines Act
- KU7.** the duties and responsibilities of workmen under the Atomic Energy Regulatory Board (AERB) Act
- KU8.** the use and functioning of equipment, such as metering, pumps, centrifugal pumps, diaphragm pumps, etc

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** make note of work-related observations
- GS2.** interpret the relevant drawings, sketches, symbols, measurements and Original Equipment Manufacturer (OEM) manuals
- GS3.** use the online Enterprise Resource Planning (ERP) systems under guidance of the supervisor
- GS4.** communicate politely and professionally
- GS5.** use effective questioning techniques to extract maximum information about different tasks
- GS6.** plan daily tasks appropriately to ensure their timely completion
- GS7.** undertake different activities as per the applicable task lists and schedules with minimum supervision
- GS8.** make appropriate decisions concerning the area of work
- GS9.** follow the 5s principles at work
- GS10.** use reasoning skills to identify and resolve basic problems at work
- GS11.** assist in identifying and implementing appropriate solutions to work-related problems
- GS12.** follow the organizational protocol during emergencies

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Assist in installing processing machinery and conveying equipment</i>	7	10	-	4
PC1. prepare the mechanical components for the maintenance of rare earth extraction plant equipment, e.g. grinding, crystallization, evaporators, centrifuge, flash drier setups, filter press, mixer settlers, etc	4	6	-	2
PC2. assemble the relevant equipment in chemical plant processing, material handling, etc	3	4	-	2
<i>Maintain the rare earth extraction plant equipment</i>	23	40	-	16
PC3. diagnose and repair faults in mechanical systems, such as gears, pumps, agitators, valves, mixer settler fans, etc	3	5	-	2
PC4. ensure proper protective enclosures are provided for rotating parts, such as drives, belts etc	3	3	-	2
PC5. follow the appropriate safety measures, not performing maintenance of a working machine	3	4	-	3
PC6. conduct appropriate tests (e.g. load testing, vibration testing, etc.) on the equipment after repair and maintenance to ensure everything is working correctly and safely	4	10	-	3
PC7. carry out predictive, preventive and breakdown maintenance for the processing, conveying and other equipment, such as fans, pumps, valves pipelines, gearbox, etc.	4	10	-	2
PC8. follow the maintenance schedule and standard procedure for equipment maintenance	3	4	-	2
PC9. conduct the rubber line spark test	3	4	-	2
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	MIN/N3226
NOS Name	Maintain and repair the rare earth extraction plant equipment
Sector	Mining
Sub-Sector	Engineering Services
Occupation	Mechanical Services
NSQF Level	3.0
Credits	2
Version	1.0
Last Reviewed Date	15/03/2024
Next Review Date	14/03/2027
NSQC Clearance Date	15/03/2024

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MIN/N1810: Follow the applicable health, safety and environmental guidelines (Placer Mining)

Description

This unit is about following the applicable health, safety and environmental guidelines.

Scope

The scope covers the following :

- Maintain personal health and safety
- Assist in hazard management
- Work on equipment and machinery
- Check the first aid box, firefighting and safety equipment
- Report incidents

Elements and Performance Criteria

Maintain personal health and safety

To be competent, the user/individual on the job must be able to:

- PC1.** follow the appropriate hand hygiene measures, i.e., washing hands with EDTA/Soda Ash (Ethylene Diamine Tetra Acetate/Sodium Carbonate), soap and water or alcohol-based hand sanitizers
- PC2.** ensure the plants are provided with a dilution ventilation system to avoid the undue accumulation of chemical vapours in working environments
- PC3.** clean and disinfect the regularly used objects, such as mobile phones, wireless communication systems keys, wallets and work passes
- PC4.** follow the Safety Management Plan (SMP) and Emergency Management Plan (EMP)
- PC5.** follow sanitation measures in the work area, e.g., avoiding eating, drinking, chewing and smoking
- PC6.** use the appropriate PPE, e.g. using an appropriate mask to cover the mouth and nose
- PC7.** identify six directional hazards at the workplace and take decisions accordingly
- PC8.** follow appropriate safety measures to avoid exposure to radiation, e.g. limiting stay in the radiation area and avoiding working with open wounds or exposed body parts in radiation areas
- PC9.** undertake the Take-5 (Personal Risk Assessment) before the commencement of any work
- PC10.** lift objects carefully maintaining the correct body posture to avoid injury
- PC11.** use self-rescue apparatus appropriately when required
- PC12.** work in the assigned work zone in the designated work area
- PC13.** follow the recommended safety measures while working at haul roads, heights, dumps, sump areas, stockyards, near moving parts, etc
- PC14.** wear Thermo luminescent Dosimeter (TLD) Badge for personal monitoring in rare earth extraction plant
- PC15.** use caution signs as applicable and obey safety signs, stickers, and tags

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PC16. identify the environmental impact of operations related to mining, mineral separation plant and rare earth extraction plant and take steps to reduce the impact

PC17. check for contamination of sensitive body parts, e.g. face, eyes etc., and seek immediate medical help

Assist in hazard management

To be competent, the user/individual on the job must be able to:

PC18. label and store chemicals appropriately

PC19. display the MSDS for the hazardous chemicals in storage and handling areas

PC20. educate the personnel at the workplace about the safety requirements and follow the applicable safety guidelines

PC21. check the body and clothing for contamination at the end of radioactive work, and decontaminate and take a shower at the decontamination centre

PC22. maintain regular and open communication with co-workers and the employer

Work on equipment and machinery

To be competent, the user/individual on the job must be able to:

PC23. check and ensure every instrument, apparatus and equipment is AERB approved

PC24. operate the appropriate tools, equipment and machinery with proper training and authorization

Check the first aid box, firefighting, and safety equipment

To be competent, the user/individual on the job must be able to:

PC25. check the availability of first aid kits and fire extinguishers, and ensure they are easily accessible

PC26. ensure the availability of emergency showers and eye washers in the chemical handling, unloading and pumping areas

PC27. monitor the work environment at appropriate intervals for hydrogen chloride and hydrocarbons using portable monitors

PC28. operate the appropriate type of fire extinguishers to control different types of fire at the worksite

Report incidents

To be competent, the user/individual on the job must be able to:

PC29. identify and report potential hazards/ unsafe conditions and practices

PC30. follow the appropriate emergency procedure during emergencies, such as power failures, cyclones, etc.

PC31. report the emergencies and any safety incidents as per the organizational procedure

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. different types of chemical plants and their operations

KU2. the importance of time-keeping, discipline and punctuality

KU3. various chemical processes carried out in a rare earth extraction plant

KU4. the impact of violating applicable safety procedures

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- KU5.** the relevant quality norms
- KU6.** different types of individual rare earth chemical processes and the use of associated equipment
- KU7.** different types of effluent treatment processes and the use of associated equipment
- KU8.** the use of different water purifying and waste treatment agents
- KU9.** the hazards and the appropriate safety measures in acid/ alkali handling activities and usage of relevant PPE
- KU10.** hazards and safety aspects involved in acid/ alkali handling activities and usage of relevant PPE
- KU11.** the internal safety organization and the role of the safety committee, workforce inspector and AERB, etc
- KU12.** the importance of sensitization towards different genders and persons with disabilities (PWD)
- KU13.** the Importance of following infection control policies, '5-S' practices, and waste management as prescribed by regulatory authorities like the Atomic Energy Regulatory Board (AERB), United Nations Development Group (UNDG), etc.
- KU14.** the importance of water/material/energy conservation and management
- KU15.** the importance of Social Management Plan (SMP) and Environment Management Plan (EMP) prepared by the organization, as per DGMS guideline
- KU16.** the importance of wearing own TLD or pocket dosimeter (to be obtained from the Health Physics Unit) in the active area
- KU17.** the importance of staying alert and attentive to the work surroundings
- KU18.** the importance of staying in the assigned work zone in the designated work area
- KU19.** the benefits of making the safety rules part of new employee orientation
- KU20.** the importance of contacting the supervisor or manager for instruction, guidance, or training when required

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** complete appropriate documentation concerning work
- GS2.** make appropriate decisions concerning the area of work with effective reasoning
- GS3.** read and explain manuals, health, and safety instructions, memos, etc
- GS4.** plan and organize work effectively
- GS5.** detect problems at work and use reasoning skills to resolve them promptly and effectively
- GS6.** complete the assigned tasks timely

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Maintain personal health and safety</i>	16	24	-	11
PC1. follow the appropriate hand hygiene measures, i.e., washing hands with EDTA/Soda Ash (Ethylene Diamine Tetra Acetate/Sodium Carbonate), soap and water or alcohol-based hand sanitizers	1	2	-	1
PC2. ensure the plants are provided with a dilution ventilation system to avoid the undue accumulation of chemical vapours in working environments	1	1	-	1
PC3. clean and disinfect the regularly used objects, such as mobile phones, wireless communication systems keys, wallets and work passes	1	1	-	-
PC4. follow the Safety Management Plan (SMP) and Emergency Management Plan (EMP)	1	2	-	-
PC5. follow sanitation measures in the work area, e.g., avoiding eating, drinking, chewing and smoking	1	2	-	-
PC6. use the appropriate PPE, e.g. using an appropriate mask to cover the mouth and nose	1	2	-	-
PC7. identify six directional hazards at the workplace and take decisions accordingly	1	2	-	1
PC8. follow appropriate safety measures to avoid exposure to radiation, e.g. limiting stay in the radiation area and avoiding working with open wounds or exposed body parts in radiation areas	1	2	-	1
PC9. undertake the Take-5 (Personal Risk Assessment) before the commencement of any work	1	2	-	1
PC10. lift objects carefully maintaining the correct body posture to avoid injury	-	1	-	-
PC11. use self-rescue apparatus appropriately when required	1	1	-	1

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. work in the assigned work zone in the designated work area	1	1	-	1
PC13. follow the recommended safety measures while working at haul roads, heights, dumps, sump areas, stockyards, near moving parts, etc	1	1	-	1
PC14. wear Thermo luminescent Dosimeter (TLD) Badge for personal monitoring in rare earth extraction plant	1	1	-	1
PC15. use caution signs as applicable and obey safety signs, stickers, and tags	1	1	-	-
PC16. identify the environmental impact of operations related to mining, mineral separation plant and rare earth extraction plant and take steps to reduce the impact	1	1	-	1
PC17. check for contamination of sensitive body parts, e.g. face, eyes etc., and seek immediate medical help	1	1	-	1
<i>Assist in hazard management</i>	5	8	-	4
PC18. label and store chemicals appropriately	1	1	-	1
PC19. display the MSDS for the hazardous chemicals in storage and handling areas	1	2	-	1
PC20. educate the personnel at the workplace about the safety requirements and follow the applicable safety guidelines	1	1	-	-
PC21. check the body and clothing for contamination at the end of radioactive work, and decontaminate and take a shower at the decontamination centre	1	2	-	1
PC22. maintain regular and open communication with co-workers and the employer	1	2	-	1
<i>Work on equipment and machinery</i>	2	4	-	1
PC23. check and ensure every instrument, apparatus and equipment is AERB approved	1	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC24. operate the appropriate tools, equipment and machinery with proper training and authorization	1	2	-	1
<i>Check the first aid box, firefighting, and safety equipment</i>	4	10	-	2
PC25. check the availability of first aid kits and fire extinguishers, and ensure they are easily accessible	1	3	-	-
PC26. ensure the availability of emergency showers and eye washers in the chemical handling, unloading and pumping areas	1	2	-	-
PC27. monitor the work environment at appropriate intervals for hydrogen chloride and hydrocarbons using portable monitors	1	2	-	1
PC28. operate the appropriate type of fire extinguishers to control different types of fire at the worksite	1	3	-	1
<i>Report incidents</i>	3	4	-	2
PC29. identify and report potential hazards/ unsafe conditions and practices	1	2	-	-
PC30. follow the appropriate emergency procedure during emergencies, such as power failures, cyclones, etc.	1	1	-	1
PC31. report the emergencies and any safety incidents as per the organizational procedure	1	1	-	1
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	MIN/N1810
NOS Name	Follow the applicable health, safety and environmental guidelines (Placer Mining)
Sector	Mining
Sub-Sector	Mining Operation
Occupation	Placer Mining
NSQF Level	4
Credits	1
Version	1.0
Last Reviewed Date	15/03/2024
Next Review Date	14/03/2027
NSQC Clearance Date	15/03/2024

Qualification Pack

MIN/N1811: Work collaboratively and effectively in a team

Description

This unit is about the importance of teamwork in the workplace.

Scope

The scope covers the following :

- Work effectively with co-workers
- Communicate effectively with co-workers
- Practice inclusiveness at work

Elements and Performance Criteria

Work effectively with co-workers

To be competent, the user/individual on the job must be able to:

- PC1.** identify own direct and indirect responsibilities, e.g. duties concerning team members
- PC2.** undertake appropriate activities to promote cooperation and good relationships with co-workers
- PC3.** mentor and assist subordinates in the execution of their work responsibilities
- PC4.** identify possible disruptions to work through coordination with the relevant stakeholders and take appropriate preventive measures
- PC5.** follow the recommended practices to avoid and resolve conflicts at work
- PC6.** interact effectively and cooperate with co-workers
- PC7.** manage workplace conflicts, ensuring no impact on work
- PC8.** identify and take responsibility for own mistakes
- PC9.** follow the organizational policies and procedures
- PC10.** follow the appropriate productivity enhancement measures, e.g. Quality circle, 5s, TPM, Six Sigma, JIT, etc.
- PC11.** exchange feedback with peers and other team members to improve the quality of work

Communicate effectively with co-workers

To be competent, the user/individual on the job must be able to:

- PC12.** follow the organizational policy for the efficient and timely dissemination of information to the authorized personnel
- PC13.** communicate clearly and politely to ensure effective communication with co-workers
- PC14.** practice active listening to understand the conveyed information and instructions
- PC15.** practice openness in work conversations, giving undivided attention to enhance the quality of workplace communication
- PC16.** maintain the professional standard of communication, e.g. avoiding the use of offensive language at work

Practice inclusiveness at work

To be competent, the user/individual on the job must be able to:

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- PC17.** empathize with Persons with Disabilities (PwD)
- PC18.** practice gender equality at work
- PC19.** maintain respectful behaviour and etiquette at work
- PC20.** listen to and acknowledge the feelings, concerns, opinions, and ideas of others

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the importance of teamwork and the consequence of poor teamwork on project outcomes, timelines, safety, etc
- KU2.** the duties and rights of workers
- KU3.** the importance of creating a healthy and cooperative work environment
- KU4.** the importance of proper and effective communication and the adverse effects of poor communication on quality, timeliness, safety, etc
- KU5.** different modes of communication used at work
- KU6.** the importance and need of supporting co-workers facing problems for the smooth functioning of work
- KU7.** the fundamental concept of gender equality
- KU8.** how to recognize and be sensitive to issues of disability, culture and gender
- KU9.** how to show empathy to PwD appropriately
- KU10.** legislation, policies, and procedures relating to gender sensitivity and cultural diversity including their impact on the area of operation
- KU11.** the appropriate behaviour to be practised with coworkers
- KU12.** the appropriate communication skills and leadership qualities
- KU13.** how to be open to listening to and understanding the other person's point of view

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** complete appropriate documentation concerning work
- GS2.** make appropriate decisions concerning the area of work with effective reasoning
- GS3.** read and explain manuals, health, and safety instructions, memos, etc
- GS4.** plan and organize work effectively
- GS5.** detect problems at work and use reasoning skills to resolve them promptly and effectively
- GS6.** complete the assigned tasks timely

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Work effectively with co-workers</i>	15	30	-	10
PC1. identify own direct and indirect responsibilities, e.g. duties concerning team members	3	5	-	3
PC2. undertake appropriate activities to promote cooperation and good relationships with co-workers	1	2	-	1
PC3. mentor and assist subordinates in the execution of their work responsibilities	1	2	-	1
PC4. identify possible disruptions to work through coordination with the relevant stakeholders and take appropriate preventive measures	3	4	-	2
PC5. follow the recommended practices to avoid and resolve conflicts at work	1	2	-	1
PC6. interact effectively and cooperate with co-workers	1	1	-	-
PC7. manage workplace conflicts, ensuring no impact on work	1	1	-	-
PC8. identify and take responsibility for own mistakes	1	3	-	1
PC9. follow the organizational policies and procedures	1	3	-	-
PC10. follow the appropriate productivity enhancement measures, e.g. Quality circle, 5s, TPM, Six Sigma, JIT, etc.	1	4	-	1
PC11. exchange feedback with peers and other team members to improve the quality of work	1	3	-	-
<i>Communicate effectively with co-workers</i>	9	10	-	5
PC12. follow the organizational policy for the efficient and timely dissemination of information to the authorized personnel	2	3	-	2

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. communicate clearly and politely to ensure effective communication with co-workers	1	2	-	-
PC14. practice active listening to understand the conveyed information and instructions	2	1	-	1
PC15. practice openness in work conversations, giving undivided attention to enhance the quality of workplace communication	2	2	-	1
PC16. maintain the professional standard of communication, e.g. avoiding the use of offensive language at work	2	2	-	1
<i>Practice inclusiveness at work</i>	6	10	-	5
PC17. empathize with Persons with Disabilities (PwD)	2	2	-	2
PC18. practice gender equality at work	2	2	-	2
PC19. maintain respectful behaviour and etiquette at work	1	3	-	1
PC20. listen to and acknowledge the feelings, concerns, opinions, and ideas of others	1	3	-	-
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	MIN/N1811
NOS Name	Work collaboratively and effectively in a team
Sector	Mining
Sub-Sector	Mining Operation
Occupation	Placer Mining
NSQF Level	4
Credits	1
Version	1.0
Last Reviewed Date	15/03/2024
Next Review Date	14/03/2027
NSQC Clearance Date	15/03/2024

Qualification Pack

DGT/VSQ/N0101: Employability Skills (30 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. understand the significance of employability skills in meeting the job requirements

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC4. speak with others using some basic English phrases or sentences

Communication Skills

To be competent, the user/individual on the job must be able to:

PC5. follow good manners while communicating with others

PC6. work with others in a team

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Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC7. communicate and behave appropriately with all genders and PwD

PC8. report any issues related to sexual harassment

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC9. use various financial products and services safely and securely

PC10. calculate income, expenses, savings etc.

PC11. approach the concerned authorities for any exploitation as per legal rights and laws

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC12. operate digital devices and use its features and applications securely and safely

PC13. use internet and social media platforms securely and safely

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC14. identify and assess opportunities for potential business

PC15. identify sources for arranging money and associated financial and legal challenges

Customer Service

To be competent, the user/individual on the job must be able to:

PC16. identify different types of customers

PC17. identify customer needs and address them appropriately

PC18. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC19. create a basic biodata

PC20. search for suitable jobs and apply

PC21. identify and register apprenticeship opportunities as per requirement

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use basic spoken English language

KU6. Do and dont of effective communication

KU7. inclusivity and its importance

KU8. different types of disabilities and appropriate communication and behaviour towards PwD

KU9. different types of financial products and services

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- KU10.** how to compute income and expenses
- KU11.** importance of maintaining safety and security in financial transactions
- KU12.** different legal rights and laws
- KU13.** how to operate digital devices and applications safely and securely
- KU14.** ways to identify business opportunities
- KU15.** types of customers and their needs
- KU16.** how to apply for a job and prepare for an interview
- KU17.** apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively using appropriate language
- GS2.** behave politely and appropriately with all
- GS3.** perform basic calculations
- GS4.** solve problems effectively
- GS5.** be careful and attentive at work
- GS6.** use time effectively
- GS7.** maintain hygiene and sanitisation to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. understand the significance of employability skills in meeting the job requirements	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	1	3	-	-
PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC4. speak with others using some basic English phrases or sentences	-	-	-	-
<i>Communication Skills</i>	1	1	-	-
PC5. follow good manners while communicating with others	-	-	-	-
PC6. work with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	1	-	-
PC7. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC8. report any issues related to sexual harassment	-	-	-	-
<i>Financial and Legal Literacy</i>	3	4	-	-
PC9. use various financial products and services safely and securely	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. calculate income, expenses, savings etc.	-	-	-	-
PC11. approach the concerned authorities for any exploitation as per legal rights and laws	-	-	-	-
<i>Essential Digital Skills</i>	4	6	-	-
PC12. operate digital devices and use its features and applications securely and safely	-	-	-	-
PC13. use internet and social media platforms securely and safely	-	-	-	-
<i>Entrepreneurship</i>	3	5	-	-
PC14. identify and assess opportunities for potential business	-	-	-	-
PC15. identify sources for arranging money and associated financial and legal challenges	-	-	-	-
<i>Customer Service</i>	2	2	-	-
PC16. identify different types of customers	-	-	-	-
PC17. identify customer needs and address them appropriately	-	-	-	-
PC18. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	1	3	-	-
PC19. create a basic biodata	-	-	-	-
PC20. search for suitable jobs and apply	-	-	-	-
PC21. identify and register apprenticeship opportunities as per requirement	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0101
NOS Name	Employability Skills (30 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	2
Credits	1
Version	1.0
Last Reviewed Date	18/02/2026
Next Review Date	18/02/2029
NSQC Clearance Date	18/02/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Element/ Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each Element/ PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
6. To pass the Qualification Pack assessment, every trainee should score the Recommended Pass % aggregate for the QP.
7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

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Minimum Aggregate Passing % at QP Level : 70

(**Please note:** Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
MIN/N3224.Maintain and repair the dredging and SMPs unit equipment	30	50	0	20	100	50
MIN/N3225.Maintain and repair the mineral separation plant equipment	30	50	0	20	100	20
MIN/N3226.Maintain and repair the rare earth extraction plant equipment	30	50	0	20	100	15
MIN/N1810.Follow the applicable health, safety and environmental guidelines (Placer Mining)	30	50	0	20	100	5
MIN/N1811.Work collaboratively and effectively in a team	30	50	0	20	100	5
DGT/VSQ/N0101.Employability Skills (30 Hours)	20	30	-	-	50	5
Total	170	280	-	100	550	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.